

Axiom Switch Universe Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 35-45 min

Category	Mathematics	Time	35-45 min
Difficulty	Extended	ID	axiom-switch-universe

Big idea: Axioms create mathematical worlds.

One-Minute Teacher Brief

Students explore how changing assumptions changes mathematical truth.

Student Prompt

Inspect Axiom Switch Universe Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Axiom Switch Universe Stimulus A	Student-facing stimulus 1: A geometry rule change where parallel lines behave differently. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Axiom Switch Universe Stimulus B	Student-facing stimulus 2: A made-up arithmetic system with a changed operation rule. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Axiom Switch Universe Reveal / Twist	Student-facing stimulus 3: A reflection on whether certainty depends on starting assumptions. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Axioms create mathematical worlds.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Axiom Switch Universe Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Axiom Switch Universe Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Are mathematical truths absolute?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how axioms create mathematical worlds.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Are mathematical truths absolute?

Setup Checklist

- Prepare a starter stimulus using: A geometry rule change where parallel lines behave differently.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- A geometry rule change where parallel lines behave differently.
- A made-up arithmetic system with a changed operation rule.
- A reflection on whether certainty depends on starting assumptions.

Resources To Use

- Resource pack: <resources/axiom-switch-universe/index.html>
- Card deck CSV: <resources/axiom-switch-universe/cards.csv>
- Visual prompt: <resources/axiom-switch-universe/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Give groups different geometry worlds: exactly one parallel line, no parallel lines, or many parallel lines through a point not on a line.
2. Students draw what triangles, lines and maps might be like in their world.
3. Each group states one theorem that may change.
4. Discuss usefulness, consistency and truth.

Debrief Moves

- Knowledge question: Are mathematical truths absolute?
- If different axiom systems are possible, what makes one useful?
- Is consistency enough for mathematical knowledge?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...